# SmartTea-LK C04: Formulas and Methods

AI-Based Tea Plantation Resource Optimization and Intervention Planning · Dasith Sathpura · 21 October 2026

## Overview

The system turns a soil reading and a weather forecast into one recommendation through three tiers, and each tier uses its own formulas. The machine-learning models learn their relationships from data; every other formula here is written out explicitly.

| Tier | What it decides | Formulas used | Section |
|---|---|---|---|
| 1. Prediction | Yield of a plucking round; % of fertilizer lost to runoff | Gradient-boosted trees, training loss, monotone rule | 1 |
| 1. Prediction (check) | How good the models are | $R^2$, MAE, RMSE, accuracy, precision, recall, F1 | 2 |
| 2. Optimizer | How much nitrogen and water, which product | Effective N, soil-N response, 90-day gain, pick rule, constraints | 3, 4 |
| 3. Action Window | When to apply in the next 4 days | Rain, soil-moisture, wind, pH and blight rules | 5 |
| Data | The simulated part of the hybrid dataset | Soil water, growth, fertilizer loss, nitrogen, blight | 6 |

All symbols and parameter values are collected in section 7.

## 1. Prediction models (Tier 1)

Both models are gradient-boosted decision trees: the yield model is CatBoost (1,947 trees) and the runoff model is XGBoost (171 trees). Each prediction is the starting value plus the small corrections of many shallow trees.

### Prediction formula

$$\hat{y}(x) = F_0 + \eta \sum_{m=1}^{M} f_m(x)$$

- x = the inputs of one case (57 for yield, 24 for runoff)
- $F_0$ = the starting value (the mean of the training target)
- $f_m$ = one decision tree; it sends x to a leaf and returns that leaf's value w
- η = learning rate = 0.03 (each tree adds only a small step)
- M = number of trees, fixed by early stopping (below)

Each new tree is fitted to the errors left by the trees before it:

$$F_m(x) = F_{m-1}(x) + \eta\, f_m(x)$$

## Training objective

Training minimises the prediction error plus a penalty on tree complexity. The penalty is what keeps the models from memorising the training data.

$$\mathcal{L} = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \sum_{m=1}^{M} \left( \gamma T_m + \tfrac{1}{2}\lambda \sum_{j=1}^{T_m} w_{mj}^2 \right)$$

- $y_i$ = actual value, $\hat{y}_i$ = prediction, n = number of training rows
- $T_m$ = number of leaves in tree m; $w_{mj}$ = value of leaf j
- γ = cost of adding a leaf; λ = L2 penalty that shrinks leaf values towards 0

With squared error, each row has gradient $g_i = \hat{y}_i - y_i$ and second derivative $h_i = 1$. The best value of a leaf, and the gain of splitting a leaf into left (L) and right (R), are then:

$$w_j^* = -\frac{\sum_{i \in j} g_i}{\sum_{i \in j} h_i + \lambda}$$

$$\text{Gain} = \frac{1}{2}\left[ \frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{(G_L + G_R)^2}{H_L + H_R + \lambda} \right] - \gamma$$

G and H are the sums of g and h on each side. A split is made only when its gain is positive.

## Early stopping

Trees are added while the error on the 2024 validation year keeps falling. Training stops after 100 trees in a row bring no improvement, and the best count M is kept. The chosen settings are then refitted on 2021–2024 with that M.

## Settings chosen on 2024 data

| Setting | Yield model (CatBoost) | Runoff model (XGBoost) |
|---|---|---|
| Learning rate η | 0.03 | 0.03 |
| Tree depth | 4 (symmetric trees) | 4 |
| L2 penalty λ | 3 | 5 |
| Minimum rows in a leaf (min child weight) | — | 5 |
| Row / column sampling per tree | — | 70 % / 60 % |
| Number of trees M | 1,947 | 171 |
| Inputs | 57 | 24 |
| Target | green leaf, kg/ha/day | % of fertilizer lost |

CatBoost builds symmetric trees (the same split across a whole tree level) and uses ordered boosting, which reduces overfitting on small data.

### Monotone rule (runoff model)

More forecast rain can never lower the predicted loss. For each forecast-rain input r (next 12, 24, 48 and 72 h), with every other input fixed:

$$r' \geq r \;\Rightarrow\; \hat{y}(x, r') \geq \hat{y}(x, r)$$

XGBoost enforces this inside every split. A check on all 170 test rows found 0 violations.

## 2. Evaluation formulas

On the unseen test year 2025, the yield model explains 91 % of yield variation ($R^2$ 0.91) and the runoff model classifies high/low risk with F1 0.82. All scores are computed once on data the model never saw in training.

### For numbers (yield, runoff amount)

**$R^2$, the share of variation explained** (1 = perfect, 0 = no better than always predicting the average $\bar{y}$):

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2}$$

**MAE, the average size of an error**, in the target's own units:

$$\mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

**RMSE**, like MAE but large errors count more:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

### For the high/low runoff-risk decision

A case is "high risk" when the loss is above 23.9 %, the median loss of 2021–2023, fixed before testing. TP = high-risk cases predicted high, FP = low-risk cases predicted high, FN = high-risk cases predicted low, TN = low-risk cases predicted low.

$$\mathrm{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\mathrm{Precision} = \frac{TP}{TP + FP} \qquad \mathrm{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2 \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$$

### Results

| Model | Measure | Test year 2025 | New blocks and a new year |
|---|---|---|---|
| Yield (CatBoost) | $R^2$ | 0.91 | 0.80 |
| Yield (CatBoost) | MAE / RMSE | 1.63 / 2.15 kg/ha/day | — |
| Runoff amount (XGBoost) | $R^2$ | 0.57 | 0.44 |
| Runoff amount (XGBoost) | MAE / RMSE | 9.9 / 13.4 percentage points | — |
| Runoff risk (XGBoost) | Accuracy | 0.81 (guessing: 0.54) | 0.79 |
| Runoff risk (XGBoost) | Precision / Recall | 0.80 / 0.85 | — |
| Runoff risk (XGBoost) | F1 | 0.82 | 0.81 |

**How the tests were split.** Settings were chosen on 2024 only and 2025 was scored once. "New blocks and a new year" trains on the other blocks' 2021–2023 rows and predicts the held-out blocks' 2025 rows, in 5 folds of 2 blocks per estate.

## 3. Effective nitrogen and the soil-N response

Each kg of nitrogen that is not washed away raises the soil-N reading by 0.27 ppm, and the effect fades with a 45-day time constant. This model, learned from the 2021–2023 sensor data, carries a fertilizer dose's effect into later plucking rounds.

### Nitrogen in a dose

$$N = D \times c_p$$

D = mixture applied (kg/ha); $c_p$ = nitrogen share of product p: NPK_BAL 0.21, NPK_HN 0.27, UREA 0.46.

### Effective nitrogen

The runoff model predicts the % lost, L, under the forecast at decision time:

$$N_{\text{eff}} = N \times \left(1 - \frac{L}{100}\right)$$

### Soil-N response to one application

The extra soil N measured $\ell$ days after an application:

$$\Delta S(\ell) = \alpha \, N_{\text{eff}} \, e^{-\ell/\tau}, \qquad \ell \geq 1$$

- α = 0.2708 ppm per kg of N_eff (learned; the data generator's true value is 0.35)
- τ = 45 days (learned): after 45 days 37 % of the effect remains, after 90 days 14 %

### How α and τ were fitted

The 7-day mean of the soil-N sensor, S, for block b on day t is modelled as a block level, a season level and the sum of all past applications:

$$S_{b,t} = a_b + s_{z(b),\,m(t)} + \alpha \sum_{e:\,t_e<t} N_{\text{eff},e}\; e^{-(t-t_e)/\tau}$$

- a_b = level of block b; s = level of the block's zone z in calendar month m (season)
- the sum runs over the block's applications e on days $t_e$

For each τ in {10, 15, 20, 30, 45, 60, 90, 120, 180, 270, 365} days, α and the levels are found by least squares on 2021–2023. The τ with the best 2024 fit is kept.

| Model | $R^2$ on 2024 | $R^2$ on 2025 |
|---|---|---|
| Block and season levels only | 0.46 | — |
| With the fertilizer term (τ = 45 days) | 0.54 | 0.41 |

### Soil N before a future round

The yield model uses the 7-day mean soil N before each round. For a round that starts s days after the decision, an application made a days earlier contributes the mean of its decay over those 7 days:

$$W(a,s) = \frac{1}{7}\sum_{\ell=s-7}^{s-1} \mathbf{1}[a+\ell \ge 1]\; e^{-(a+\ell)/\tau}$$

The soil N fed to the yield model for round j, starting $s_j$ days ahead, is then:

$$S_j = S_{\text{now}} + \alpha \sum_e N_{\text{eff},e}\,[W(a_e,s_j) - W(a_e,0)] + \alpha\, N_{\text{eff,new}}\, W(0,s_j)$$

- S_now = the latest 7-day reading
- the middle term = earlier applications that keep fading
- the last term = the new dose being considered
- $S_j$ is capped at 87.2 ppm, the highest value in the training data, because tree models cannot extrapolate

## 4. Optimizer (Tier 2, NSGA-II)

At the start of each plucking round the optimizer chooses a nitrogen dose and an irrigation amount, and recommends a dose only if each kg of nitrogen adds at least 3.5 kg of green leaf over the next 90 days.

### Decision variables and limits

$$N \in \{0\} \cup [N_{lo}, N_{hi}], \qquad I \in [0, I_{hi}]$$

N = nitrogen applied now (kg N/ha); I = irrigation this round (mm). A value below N_lo means no application.

**Nitrogen bounds** (operational rules intersected with the training range):

$$N_{lo} = 28.4, \qquad N_{hi} = \min\left(113.4,\; C_z - N_{\text{year}},\; 156.9 - N_{60},\; 200 - N_{90}\right)$$

- C_z = annual ceiling of zone z: 330 kg N/ha up-country, 300 mid- and low-country

- N_year, $N_{60}$, $N_{90}$ = nitrogen already applied this year and inside the round's 60- and 90-day windows
- 28.4 and 113.4 = smallest and largest single dose in the training data; 156.9 and 200 = training maxima of the 60- and 90-day totals
- N_hi = 0 if the last application was under 45 days ago, or pruning under 30 days ago

**Irrigation bound** (only on irrigated blocks, and only when the soil is drier than the trigger):

$$I_{hi} = \min\left(25,\ 61.9,\ \frac{V_{\text{left}}}{A \times 10\,000}\right) \quad \text{if } \theta < WP + 0.35\,(FC - WP),\ \text{else } 0$$

θ = soil moisture (%); FC, WP = field capacity and wilting point; V_left = water left in the estate's 3,000,000 L monthly budget; A = block area (ha); 1 mm on 1 ha = 10,000 L.

### Product and dose

For each product p that can deliver N within 100–400 kg/ha of mixture, the dose is rounded down to whole 5 kg, and the product with the lowest predicted runoff loss $L_p$ is chosen:

$$D_p = 5\left\lfloor \frac{N}{5\,c_p} \right\rfloor, \qquad p^* = \arg\min_p L_p$$

### Yield over the 90-day horizon

With rounds of k days, the horizon has J rounds and H days:

$$J = \left\lceil \frac{90}{k} \right\rceil, \qquad H = J\,k$$

For each round j = 0 … J−1 the yield model predicts $\hat{y}_j$, using the soil N $S_j$ from section 3 and these updated inputs:

$$N_{60,j} = N_{60,j}^{\text{prior}} + N \cdot \mathbf{1}[(j+1)k \le 60], \qquad N_{90,j} = N_{90,j}^{\text{prior}} + N \cdot \mathbf{1}[(j+1)k \le 90]$$

$$\text{days since fertilizer}_j = \begin{cases} (j+1)k & \text{if } N > 0 \\ \min\left(365,\ d_0 + (j+1)k\right) & \text{otherwise} \end{cases}$$

$d_0$ = days since the last application at decision time. Irrigation enters round 0 only. Rounds after the first use the current round's other conditions, because future weather is unknown, so only differences between plans are meaningful. The plan's yield is the average over the horizon:

$$\bar{y}(N, I) = \frac{1}{J}\sum_{j=0}^{J-1} \hat{y}_j(N, I)$$

### Objectives

$$\max\ \bar{y}(N, I), \qquad \min\ N, \qquad \min\ I$$

### How NSGA-II searches

- **Population:** 40 plans, evolved for 40 generations (seed 0). The first population always contains no action, the smallest dose and the largest dose.
- **Dominance:** plan A dominates plan B when A is at least as good on every objective and better on at least one. Plans no other plan dominates form the Pareto front.
- **Crowding distance:** keeps the front spread out. For plan i, sorted along each objective f:

$$d_i = \sum_f \frac{f_{i+1} - f_{i-1}}{f_{\max} - f_{\min}}$$

- **Check:** on 40 test plans, NSGA-II reached 100 % of the exhaustive-search optimum and chose the same dose in 95–100 % of plans.

### Picking one plan from the Pareto front

Gain over the horizon, in kg of green leaf per ha:

$$G(N, I) = [\bar{y}(N, I) - \bar{y}(0, 0)] \times H$$

Score, using two agronomic thresholds and no cost variables:

$$\text{Score}(N, I) = G(N, I) - \text{AE}_{\min} \cdot N - R_{\min} \cdot I$$

- AE_min = 3.5 kg of green leaf each kg of N must add (decision-maker setting)
- R_min = 2.0 kg of green leaf per ha each mm of water must add

The plan with the highest score is recommended (on a tie, the smaller dose). If no plan scores above 0, the recommendation is **no action**.

### Reported efficiency

$$\text{Efficiency} = 1 - \frac{L}{100}, \qquad \text{AE} = \frac{G}{N} \ \ (\text{kg leaf per kg N})$$

## 5. Action Window (Tier 3)

The Action Window checks every working hour in the next 96 hours and, for fertilizer, picks the safe hour with the lowest predicted runoff. Field work is shown as a 4-hour block starting between 06:00 and 12:00, so it ends by 16:00.

### Candidate hours

$$T = \{\, t : \text{now} < t \le \text{now} + 96\ \text{h},\ 6 \le \text{hour}(t) \le 16 - 4 \,\}$$

$R_h(t)$ = forecast rain summed over the h hours from t; $P_{12}(t)$ = highest forecast rain probability in the 12 hours from t; $U_4(t)$ = highest forecast wind speed in the 4 hours from t.

### Fertilizer

An hour is safe when all three rules hold:

$$R_{12}(t) \le 5\ \text{mm}, \qquad P_{12}(t) \le 0.30, \qquad WP + 0.25\,(FC - WP) \le \theta \le WP + 0.85\,(FC - WP)$$

Among the safe hours, the runoff model is run with that hour's forecast rain for 12, 24, 48 and 72 h, and the hour with the lowest predicted loss is chosen:

$$t^{*} = \arg\min_{t \in T_{\text{safe}}} \hat{L}(t)$$

If no hour is safe, the recommendation is to delay fertilizer, with the rule that blocked it.

### Fungicide (blister blight)

A spray is recommended when the block is above 600 m and the smoothed blight risk is above 0.35. For each past day i:

$$w_i = \frac{1}{5} \sum_{k=i-5}^{i-1} \mathbf{1}[\text{rain}_k > 1\ \text{mm}]$$

$$r_i = \text{clip}\left( w_i \left(1 - \frac{\min(\bar{s}_5, 6)}{6}\right) \times 1.7 \times \left(0.8 + \frac{RH_i}{500}\right),\ 0,\ 1 \right)$$

$$\text{Risk}_i = 0.8\,\text{Risk}_{i-1} + 0.2\,r_i$$

- $w_i$ = share of wet days among the last 5; $\bar{s}_5$ = mean sunshine hours of those 5 days; $RH_i$ = mean relative humidity (%)
- the spray hour needs $R_4(t) \le 1$ mm (4 rain-free hours) and $U_4(t) \le 4$ m/s
- at least 7 days between sprays, and no plucking for 7 days after a spray (pre-harvest interval)

Sunshine is derived from solar radiation Q (MJ/m²/day) on day-of-year d, exactly as in the training data:

$$s = \text{clip}\left( \frac{\dfrac{Q}{34.5 + 2\cos(2\pi\,(d - 80)/365)} - 0.25}{0.5} \times 12,\ 0,\ 11.5 \right)$$

### Dolomite (soil pH)

Applied when the 7-day mean soil pH is below 4.45 (target 4.5–5.5) and no dolomite was applied in the last 730 days. The hour needs $R_{24}(t) < 20$ mm.

### Irrigation

On irrigated blocks only, when the soil is below the trigger and little rain is coming:

$$\theta < \theta_{\text{trig}} = WP + 0.35\,(FC - WP) \quad \text{and} \quad R_{48} < 5\ \text{mm}$$

If 5 mm or more is forecast in 48 hours, the advice is to skip irrigation. The amount lifts the soil back above the trigger plus a typical 3.2-point step, over a 400 mm root zone with 85 % of the water reaching the roots:

$$I = \text{clip}\left(\text{round}\left(\frac{(\theta_{\text{trig}} - \theta + 3.2)}{100} \times \frac{400}{0.85}\right),\ 6,\ 25\right)\ \text{mm}$$

The amount is capped by the estate reservoir: I ≤ 3,000,000 L ÷ (A × 10,000). The hour needs $R_{12}(t) < 1$ mm, and irrigations are at least 3 days apart.

## 6. Simulation formulas behind the hybrid dataset

The management and crop-response part of the dataset was simulated day by day for each block on top of real ERA5 weather and SoilGrids soil. These formulas generated it; their coefficients are approximate placeholders still to be checked against Tea Research Institute guidance.

Symbols used here: θ = soil moisture (%), FC / WP = field capacity / wilting point (%), P = rain (mm), $ET_0$ = reference evapotranspiration (mm), Q = solar radiation (MJ/m²/day), T = mean air temperature (°C), σ = shade cover (fraction), S_N / S_P / S_K = soil N / P / K (ppm).

### Soil water balance (400 mm root zone)

$$I_{\text{cap}} = 60\,(1 - \text{slope}/100), \qquad \text{Infil} = \min(0.9P,\ I_{\text{cap}} + 0.4\max(0,\ P - I_{\text{cap}}))$$

$$k_s = \text{clip}\left(\frac{\theta - WP}{0.6\,(FC - WP)},\ 0,\ 1\right)$$

$$\theta_{t+1} = \theta_t + \frac{\text{Infil}}{400} \times 100 - \frac{0.9\,ET_0\,k_s}{400} \times 100$$

Above field capacity, half the excess drains each day: θ ← θ − 0.5 (θ − FC). θ stays between WP − 1 and FC + 6.

### Daily growth

$$g = s_z \cdot f_{\text{age}} \cdot f_{\text{cult}} \cdot E \cdot f_{\text{prune}} \cdot (1 - B) \cdot e^{\varepsilon}, \qquad \varepsilon \sim \mathcal{N}(0,\ 0.08^2)$$

- s_z = zone scale, calibrated to about 7,000 (up), 6,000 (mid) and 9,500 (low) kg green leaf/ha/year
- f_cult = 0.78 seedling; 1.00, 0.95, 1.05 for clones A, B, C
- f_age = min(1, 0.55 + 0.08 × age) under 6 years, 0.88 over 50 years, else 1
- E = the environment, smoothed over days: $E_t = 0.75\,E_{t-1} + 0.25\,e_t$
- B = blister-blight loss (below)

The daily environment multiplies seven factors:

$$e = f_T \cdot f_W \cdot f_L \cdot f_N \cdot f_K \cdot f_P \cdot f_{pH}$$

$$f_W = \text{clip}\left(\frac{\theta - WP}{0.45\,(FC - WP)},\ 0.15,\ 1\right) \quad (\times 0.92 \text{ if } \theta > FC + 2)$$

$$f_L = 0.7 + 0.3 \min\left(1,\ \frac{Q\,(1 - 0.6\,\sigma)}{15}\right)$$

$$\begin{aligned} f_N &= 0.5 + 0.5\left(1 - e^{-S_N/30}\right) \\ f_K &= 0.88 + 0.12\left(1 - e^{-S_K/70}\right) \\ f_P &= 0.95 + 0.05\left(1 - e^{-S_P/10}\right) \end{aligned}$$

The nutrient factors are Mitscherlich curves: each extra ppm helps less than the one before. f_N falls by 0.002 per ppm above 150 ppm.

$$f_T = \begin{cases} 0.2 & T < 12 \\ 0.2 + 0.8\,(T - 12)/6 & 12 \le T < 18 \\ 1 & 18 \le T \le 26 \\ 1 - 0.5\,(T - 26)/6 & 26 < T \le 32 \\ 0.5 & T > 32 \end{cases}$$

$$f_{pH} = \begin{cases} \mathrm{clip}(0.8 - 0.5\,(4.2 - pH),\ 0.5,\ 0.8) & pH < 4.2 \\ 0.8 + 0.2\,(pH - 4.2)/0.3 & 4.2 \le pH < 4.5 \\ 1 & 4.5 \le pH \le 5.5 \\ 1 - 0.3\,(pH - 5.5) & 5.5 < pH \le 6.0 \\ 0.85 & pH > 6.0 \end{cases}$$

$$f_{\text{prune}}(d) = \begin{cases} 0 & d < 45 \\ 0.35 & 45 \le d < 90 \\ 0.75 & 90 \le d < 180 \\ 0.95 & 180 \le d < 365 \\ 1.05 & 365 \le d < 730 \\ \mathrm{clip}\big(1.05 - 0.04\,\frac{d - 730}{365},\ 0.85,\ 1.05\big) & d \ge 730 \end{cases}$$

d = days since pruning.

### Fertilizer loss (runoff and volatilisation)

$r_{12}$, $r_{48}$, $r_{72}$ = actual rain (mm) in the 12, 48 and 72 hours after application:

$$P_{\text{eff}} = r_{12} + 0.5\,(r_{48} - r_{12}) + 0.2\,(r_{72} - r_{48})$$

$$\begin{aligned} \ell = {} & 0.04 + 0.62\left(1 - e^{-P_{\text{eff}}/40}\right)\left(0.7 + \frac{\text{slope}}{40}\right) \\ & + 0.10\ \mathrm{clip}\left(\frac{\theta - FC + 3}{6}, 0, 1\right) + 0.08\,u\ \mathrm{clip}\left(\frac{\theta_{\text{dry}} - \theta}{6}, 0, 1\right) + 0.02\,h \end{aligned}$$

- θ_dry = WP + 0.25 (FC − WP): dry soil loses nitrogen as gas; u = 1.5 for urea, 1 otherwise
- h = 1 when applied 11:00–15:00 on a day above 25 °C, else 0

- a random error N(0, $0.03^2$) is added and $\ell$ is kept between 0.02 and 0.90
- recorded runoff % = 100 $\ell$ + N(0, $2.5^2$), kept between 0 and 95; absorption % = 100 − runoff %

### Nutrients and pH

An application of D kg/ha with N, P and K shares adds, with 2 kg/ha = 1 ppm in the topsoil:

$$\Delta S_N = \frac{N(1-\ell)}{2} \times 0.7, \qquad \Delta S_P = \frac{P(1-\ell)}{2} \times 0.25,$$
$$\Delta S_K = \frac{K(1-\ell)}{2} \times 0.5, \qquad \Delta pH = -0.0004\, N$$

Each day, with g_rel = g / s_z (growth relative to the zone's potential):

$$S_N \leftarrow S_N + m - 0.45\, g_{\text{rel}} - 0.0025\, S_N \min\left(\frac{P}{20}, 3\right) - 0.0015\, S_N$$

m = mineralisation, 0.10 × (SoilGrids total N ÷ 4.7) ppm/day. S_P and S_K follow the same pattern with their own rates. Dolomite of D kg/ha raises pH by (0.2 + D/5000) spread over 90 days.

### Blister blight

The blight risk uses the same formula as the Action Window (wet days, sunshine, humidity; above 600 m only). Its effect on growth:

$$B = 0.30 \cdot \text{Risk} \cdot (1 - \pi)$$

π = spray protection: 0.75 after a spray, 0.30 if more than 8 mm of rain fell within 4 hours of spraying, 0 after 10 days.

## 7. Symbols and parameter values

Every fixed value used by the deployed system, with its origin: rules come from the operational constraints table, learned values from the data, and settings were chosen by the designer.

| Symbol | Meaning | Value | Unit | Origin |
|---|---|---|---|---|
| η | Learning rate of both models | 0.03 | — | Chosen on 2024 data |
| M | Trees: yield / runoff model | 1,947 / 171 | trees | Early stopping on 2024 |
| λ | L2 penalty: yield / runoff | 3 / 5 | — | Chosen on 2024 data |
| — | Tree depth: both models | 4 | levels | Chosen on 2024 data |
| α | Soil-N rise per kg effective N | 0.2708 | ppm/kg | Learned, 2021–2023 |
| τ | Soil-N fading time constant | 45 | days | Chosen on 2024 data |
| $c_p$ | N share: NPK_BAL / NPK_HN / UREA | 0.21 / 0.27 / 0.46 | — | Product records |
| AE_min | Leaf each kg N must add | 3.5 | kg leaf / kg N | Designer setting |

| Symbol | Meaning | Value | Unit | Origin |
|---|---|---|---|---|
| R_min | Leaf each mm water must add | 2.0 | kg leaf/ha per mm | Designer setting |
| H | Look-ahead horizon | 90 | days | Designer setting |
| — | NSGA-II population / generations | 40 / 40 | — | Checked against exhaustive search |
| D | Mixture per application | 100–400 | kg/ha | Operational rule |
| C_z | Annual N ceiling: up / mid, low | 330 / 300 | kg N/ha/year | Operational rule |
| — | Gap between fertilizer applications | ≥ 45 | days | Operational rule |
| — | No fertilizer after pruning | 30 | days | Operational rule |
| N_lo, N_hi | Single dose range | 28.4–113.4 | kg N/ha | Training range |
| — | Rain after fertilizer, next 12 h | ≤ 5 | mm | Operational rule |
| — | Rain probability, next 12 h | ≤ 0.30 | — | Operational rule |
| — | Fertilizer soil-moisture band | WP + 0.25–0.85 (FC − WP) | % | Operational rule |
| θ_trig | Irrigation trigger | WP + 0.35 (FC − WP) | % | Operational rule |
| — | Irrigation per event | 6–25 | mm | Operational rule |
| V | Estate water per month | 3,000,000 | L | Operational rule |
| — | Spray: rain-free time / max wind | 4 / 4 | h / m/s | Operational rule |
| — | Blight risk threshold for spraying | 0.35 | — | Data generator rule |
| — | Pre-harvest interval after a spray | 7 | days | Placeholder, check label |
| — | Dolomite trigger pH / gap | 4.45 / 730 | pH / days | Operational rule |
| — | Look-ahead and working hours | 96 h; 06:00–16:00 | — | Operational rule |
| — | High runoff-risk cut-off | 23.9 | % lost | Median of 2021–2023 |

The agronomic rules and the two designer settings (AE_min, R_min) are placeholders to be confirmed with Tea Research Institute guidance; changing them changes the recommendations, not the trained models.